

45 Hours Course | Mastering: Generative & Agentic AI

Course Duration: **45 Hours**

Level: **Beginner to Advanced**

Target Audience: **Open to ALL, B.tech, B.E, B.sc IT, M.sc It, IT Sector, Programmers, Undergraduate students, early researchers, or professionals entering into AI.**

Prerequisite Software: **Anaconda, Antigravity IDE**

→ AI, Machine Learning, and Deep Learning Fundamentals (Fast-Track)

Objective: Rapidly cover the foundational mechanics of classical AI and deep learning.

Module 1: Demystifying AI & ML Core Concepts

- **The AI Landscape:** Definitions of AI, ML, and DL. Traditional vs. AI programming
- **Types of Learning:** Supervised, Unsupervised, and Reinforcement Learning summarized.
- **Essential Vocabulary:** Features, labels, epochs, datasets, bias-variance tradeoff.

Module 2: Fundamentals of Deep Learning

- **From Neurons to Networks:** Perceptrons, weights, biases, and activation functions.
- **Network Architecture:** Layers and forward propagation.
- **Training Mechanics:** Loss functions, optimizers, and the backpropagation algorithm.

Module 3: Fully Connected and Convolutional Neural Networks (CNNs)

- **FC NNs:** Designing and training dense networks.
- **CNNs for Vision:** Convolutions, pooling, spatial feature extraction.
- **Hands-On Mini Project:** Quick image classification run-through to demonstrate network training

Module 4: Recurrent Neural Networks (RNNs)

- **Processing Sequences:** How RNNs handle time-series and text.
- **The "Memory" Problem:** Vanishing gradients and an introduction to sequential bottlenecks, leading up to why modern AI needed a new architecture.

→ **Early Generative Models & Foundation Transition**

Objective: Complete the transition from discriminative AI to generative models, laying the groundwork for modern LLMs.

Module 1: Introduction to Generative AI

- **Discriminative vs. Generative models.**
- **The generative spectrum (text, images, structured data).**
- **Autoencoders:** Basic architecture (Encoder, Bottleneck, Decoder) for dimensional reduction.

Module 2: Advanced Autoencoders & GANs

- **Variational Autoencoders (VAEs):** Probabilistic generation and latent spaces.
- **Generative Adversarial Networks (GANs):** The generator/discriminator game-theoretic structure.
- **Hands-On Mini Project:** Generating simple samples from noise using a minimal architecture.

Module 3: Introduction to the Transformer Architecture

- **The Breakthrough:** "Attention Is All You Need." Why RNNs were replaced.
- **Self-Attention Mechanism:** How words learn context from their surroundings.
- **Positional Encoding & Architecture:** The Encoder-Decoder structure that drives modern LLMs.

Module 4: Mini Transformer Demonstration

- **Hands-on Demo:** Walking through the code of a miniaturized Transformer.
- **Seeing self-attention weights and multi-head attention in action on a small sequence.**

→ Large Language Models (LLMs) & Applied Chatbots

Objective: Understand how LLMs generate text and dedicate substantial time to building a custom Q&A chatbot from scratch.

Module 1: LLM Mechanics & Decoding Strategies

- **Predicting the Next Token:** How LLMs actually "write."
- **Controlling Generation:** In-depth look at Temperature, Top-p (nucleus sampling), and Top-k.
- **Context Windows:** The limits of context stuffing and why external memory is needed.

Module 2: Building a Q&A Chatbot (Part 1: The Foundation)

- **System Design:** Starting from scratch to build an interactive chat logic flow.
- **Controlling the System:** System prompts, role-playing, and model guardrails to guide responses and prevent hallucinations.

Module 3: Building a Q&A Chatbot (Part 2: API & Memory)

- **Hands-On:** Integrating a foundation model API.
- **Managing conversation history, roles (User/Assistant), and maintaining stateless context across turns.**

Module 4: Building a Q&A Chatbot (Part 3: UI & Refinement)

- **Hands-On:** Finalizing the chatbot application (e.g., using Streamlit, Gradio, or a frontend web view in Antigravity IDE).
- Testing edge cases, system boundaries, and generation controls.

➔ **Retrieval-Augmented Generation (RAG) & Function Calling**

Objective: Overcome context limits by building RAG systems, then enhance the AI with the ability to execute tools.

Module 1: Deep Dive into RAG (Sparse v s. Dense)

- **The Need for RAG:** Grounding model responses in truth.
- **Sparse Retrievers:** Keyword-based searching (e.g., BM25/TF-IDF).
- **Dense Retrievers:** Understanding vector embeddings, semantic meaning, and vector databases.

Module 2: Hands-On: Building a Hybrid RAG Pipeline

- **Activities:** Document chunking, calculating embeddings, and implementing core retrieval mechanisms.
- Combining search results into the prompt context for the chatbot.

Module 3: Function Calling & Tool Use

- **Allowing LLMs to take action:** Defining schemas and API signatures for the model.
- Extracting structured JSON arguments and handling multi-step tool execution loops.

Module 4: Advanced RAG (Graph RAG)

- **Beyond Vector Search:** Utilizing Knowledge Graphs for highly connected, complex retrieval (Graph RAG).
- **Hands-On:** Integrating entity extraction and graph-based context into the LLM system.

→ Agentic AI, MCP, & Visual Orchestration

Objective: Build autonomous agents using standard protocols (MCP) and visual, node-based architectures (Langflow).

Module 1: Understanding the Model Context Protocol (MCP)

- **What is MCP?** Standardizing how AI systems connect to external tools, data sources, and servers.
- Server/Client architecture for AI tooling without hardcoded spaghetti code.

Module 2: Hands-On: Building an MCP-Powered Website / App

- **Project Setup:** Creating an underlying MCP server for custom data or tools.
- Building a website (UI) where the user queries securely access local/external tools via the MCP protocol.

Module 3: Introduction to Agents & Agentic AI

- **The Shift:** from stateless chatbots to goal-driven Agents.
- Memory, Planning, and Action loops (ReAct framework).
- Handling failures, retries, and multi-agent coordination.

Module 4: Building Agents with Langflow

- **Visualizing Agentic Flow:** Introduction to Langflow's interface.
- **Hands-On Capstone:** Orchestrating the tools, RAG connections, and agents built throughout the week into a single visual Langflow pipeline. Course Wrap-up & Next Steps.